

Project Design Phase – II

Technology Stack (Architecture & Stack)

Date	23 June 2026
Name	M Manosika
Project Name	Automated Network Request Management System
Maximum Marks	4 Marks

Technical Architecture Overview

The Automated Network Request Management System is developed on the ServiceNow Now Platform, providing a cloud-based environment for managing network service requests. The solution automates the complete lifecycle of a request, beginning with user submission through the Service Portal and ending with secure storage of approved request details in the database.

The architecture combines user-friendly interfaces, workflow automation, and centralized data management to improve service efficiency while reducing manual effort.



Architecture Layers

Presentation Layer (User Interface)

The presentation layer allows employees to access the Network Request Catalog through the ServiceNow Service Portal. It provides an intuitive interface for submitting requests, validating user inputs, and displaying request status updates.

Business Logic Layer

The application logic is implemented using ServiceNow Flow Designer, Business Rules, Client Scripts, and UI Policies. This layer validates request information, manages approval workflows, processes business rules, and automates task generation.

Data Management Layer

The database layer securely stores request details, user information, and workflow records. Approved requests are maintained in a dedicated custom table to support tracking, reporting, and future reference without manual data entry.

Table 1: Components and Technologies

S.No	Component	Purpose	Technology Used
1	User Interface	Provides a Service Portal where employees can submit network requests and	ServiceNow Service Portal, HTML5, CSS3, JavaScript, Bootstrap

S.No	Component	Purpose	Technology Used
		monitor request status.	
2	Client-Side Logic	Performs field validation, auto-populates user information, and controls field visibility.	Client Scripts, Catalog UI Policies, GlideUser API
3	Workflow Engine	Automates approvals, notifications, and request processing.	Flow Designer, Workflow Studio, Business Rules
4	Server-Side Processing	Handles business logic, database operations, and approval routing.	GlideRecord API, JavaScript
5	System Database	Stores user profiles, request records, and workflow information.	ServiceNow Database (MySQL/MariaDB)
6	Custom Database	Maintains approved network request records for	u_network_database

S.No	Component	Purpose	Technology Used
		tracking and reporting.	
7	File Management	Stores request attachments and supporting documents securely.	ServiceNow Attachment Storage (sys_attachment)
8	User Information API	Retrieves logged-in employee information during request creation.	GlideSystem Session APIs
9	Notification Service	Sends automated approval and status notifications through email.	ServiceNow Notification Engine (SMTP/IMAP)
10	Predictive Intelligence (Optional)	Supports intelligent task assignment.	ServiceNow Predictive Intelligence
11	Cloud Infrastructure	Provides a secure and scalable cloud environment for application deployment.	ServiceNow Platform as a Service (PaaS)

Table 2: Application Characteristics

S.No	Characteristic	Description	Technology Used
1	Responsive User Interface	Ensures the application is accessible across different screen sizes and devices.	Bootstrap Framework, AngularJS
2	Security	Protects application data through authentication, authorization, and encryption mechanisms.	AES-256 Encryption, Multi-Factor Authentication (MFA), Access Control Lists (ACLs), Role-Based Access Control
3	Scalability	Supports increasing numbers of users and requests without affecting application performance.	Multi-tier PaaS Architecture, Cloud-Based Infrastructure
4	Availability	Provides continuous system availability with high reliability and minimal downtime.	ServiceNow High Availability (AHA), Geo-Redundant Data Centers
5	Performance	Optimizes request processing, form loading, and workflow	Client-Side Caching, Indexed Database Tables, Asynchronous

S.No	Characteristic	Description	Technology Used
		execution for better user experience.	Background Processing